

Hackathon Project Phases Template

Project Title:

Logo Craft Website Development Using Diffusion Technology

Team Name:

Logo Wizards

Team Members:

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Phase-1: Brainstorming & Ideation

Objective:

- To develop and deploy LogoCraft, a user-friendly platform that utilizes Diffusion technology to generate custom logos based on textual descriptions provided by businesses.
- To democratize the logo design process, making it accessible and affordable for businesses of all sizes, regardless of their design expertise or budget

Key Points:

1.Problem StatementL:

- LogoCraft uses AI to generate custom logos from text descriptions, simplifying branding for businesses.
- This innovative approach empowers companies to create unique, professional logos that resonate with their target audience.

2.Proposed Solution:

- AI-powered logo generation: LogoCraft uses diffusion technology to create unique logos from text descriptions, simplifying design.

- Intuitive design platform: A user-friendly interface allows businesses to easily customize and refine generated logos.

3.Target Users:

- Small businesses and startups: Gain access to professional-quality logos without the high cost of designers.
- Marketing teams and entrepreneurs: Quickly create and test various logo options, streamlining branding efforts.

4.Expected Outcome:

- Democratized logo design: Makes professional-grade logos accessible to businesses of all sizes, regardless of budget.
- Streamlined branding process: Simplifies logo creation, saving time and resources for businesses and entrepreneurs.

Phase-2: Requirement Analysis

Objective:

- Define technical and functional requirements for Logo Craft Website Development.

Key Points:

1.Technical Requirements:

- Programming Languages: Python, JavaScript
- Frameworks: React.js (Frontend), Node.js (Backend)
- AI Model: Stable Diffusion for image generation
- Cloud Services: AWS/GCP for hosting and storage
- Database: Firebase / MongoDB

2.Functional Requirements:

- AI-powered logo generation based on text input
- Customization options (colors, fonts, styles)
- User authentication and saved designs
- Downloadable high-resolution logos
- Subscription-based premium features

3.Constraints & Challenges:

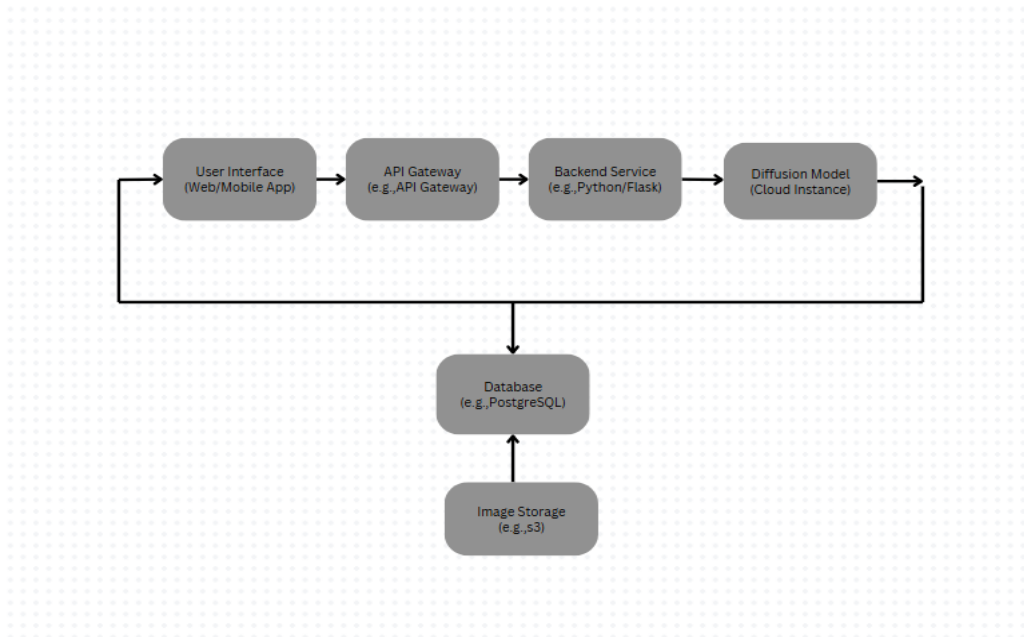
- Training and fine-tuning AI models for high-quality logo generation

- Ensuring user-friendly UI/UX for seamless interaction
 - Handling large-scale image processing efficiently
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Phase-3: Project Design

Objective:

- Create the architecture and user flow.



Key Points:

1. System Architecture Diagram:

- Cloud-based architecture with a frontend UI, backend API, AI model processing layer, and a database.

2. User Flow:

- User enters brand name and description.
- AI generates multiple logo options.
- User customizes selected logo.
- User downloads the final logo.

3.UI/UX Considerations:

- A simple and intuitive interface with drag-and-drop customization, preview mode, and responsive design.
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Phase-4: Project Planning (Agile Methodologies)

Objective:

- Break down the tasks using Agile methodologies.

Key Points:

1.Sprint Planning:

- Divide development into short sprints (AI model development, UI design, backend integration, testing, deployment).

2.Task Allocation:

- Assign roles for AI development, frontend, backend, and UI/UX.

3.Timeline & Milestones:

- Week 1-2: Research & AI model setup
- Week 3-4: Backend API and database integration
- Week 5-6: Frontend development & UI/UX implementation
- Week 7: Testing and bug fixes
- Week 8 :Deployment and the final review

Phase-5: Project Development

Objective:

- Code the project and integrate components.

Key Points:

1. Technology Stack Used:

- React.js, Node.js, Python (Stable Diffusion), Firebase, AWS

2. Development Process:

- Implement AI model for logo generation
- Develop REST API for handling user requests
- Build frontend for user interaction
- Integrate customization features and database

3. Challenges & Fixes:

- AI model optimization for better image quality
 - Enhancing real-time customization features
 - Ensuring smooth API performance
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Phase-6:Foundation & Performance Testing

Objective:

- Ensure the project works as expected.

Key Points:

1.Test Cases Executed:

- AI-generated logo quality verification
- UI/UX responsiveness testing
- API performance and response time
- User authentication and data storage validation

2.Bug Fixes & Improvements:

- Optimization of AI output, UI adjustments, and backend efficiency enhancements.

3.Final Validation:

- Ensuring all project goals and functional requirements are met.

4.Deployment (if applicable):

- Hosting on AWS/GCP with a user-accessible demo.
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Final Submission

1. **Project Report Based on the templates**
 2. **Demo Video (3-5 Minutes)**
 3. **GitHub/Code Repository Link**
 4. **Presentation**
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